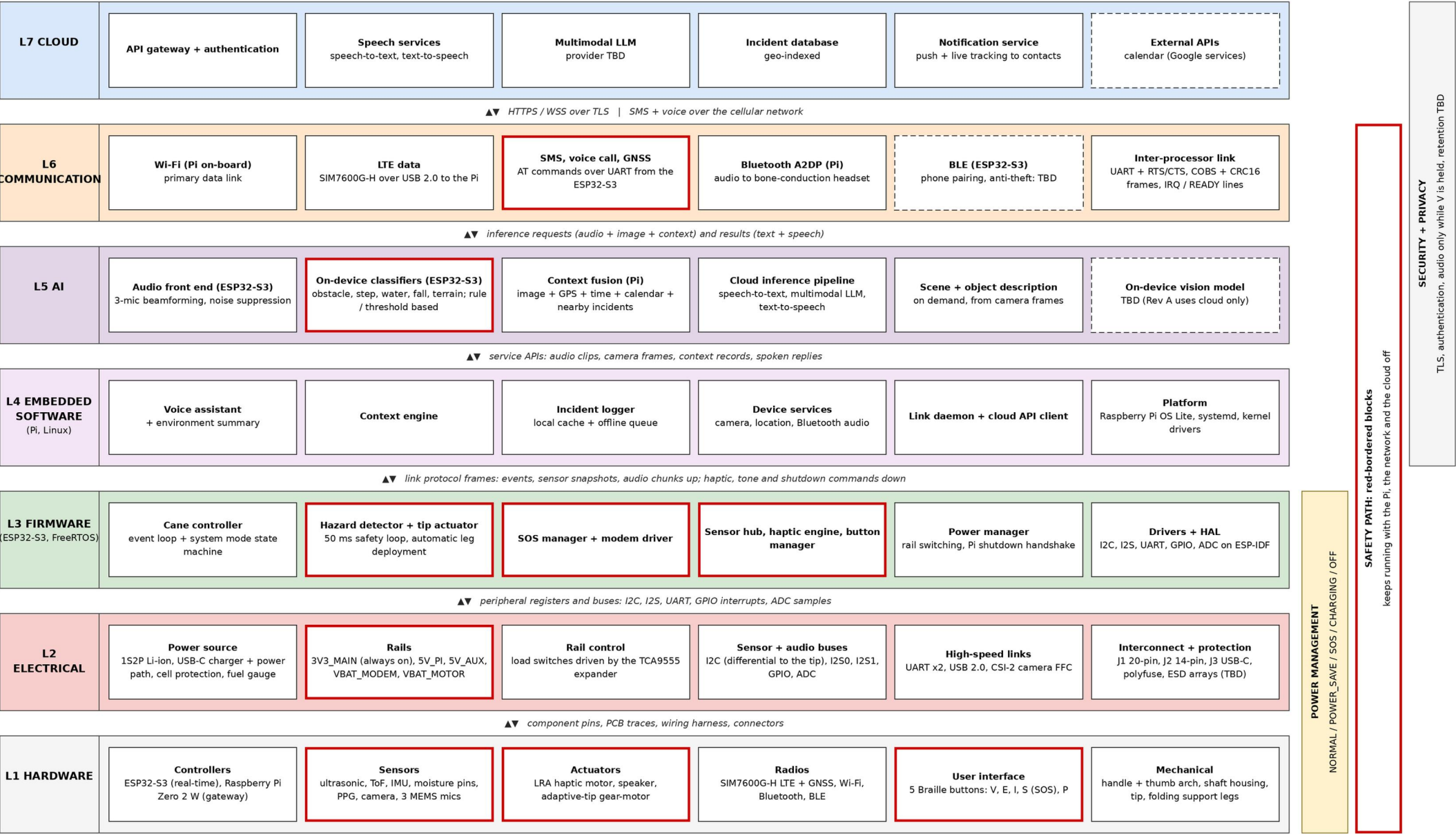


SMART INTELLIGENT CANE - HIERARCHICAL MULTI-DOMAIN SYSTEM ARCHITECTURE

Rev A (proposed). Read bottom-up: each layer uses only the layer below it, through the interface named between them.



LEGEND: band colour = domain | red 3 pt border = safety-critical | dashed border = optional or TBD | italic strip = interface between two layers | vertical bars = cross-cutting concerns and the layers they span

DIAGRAM 1 - SYSTEM-LEVEL ARCHITECTURE (Rev A, proposed)

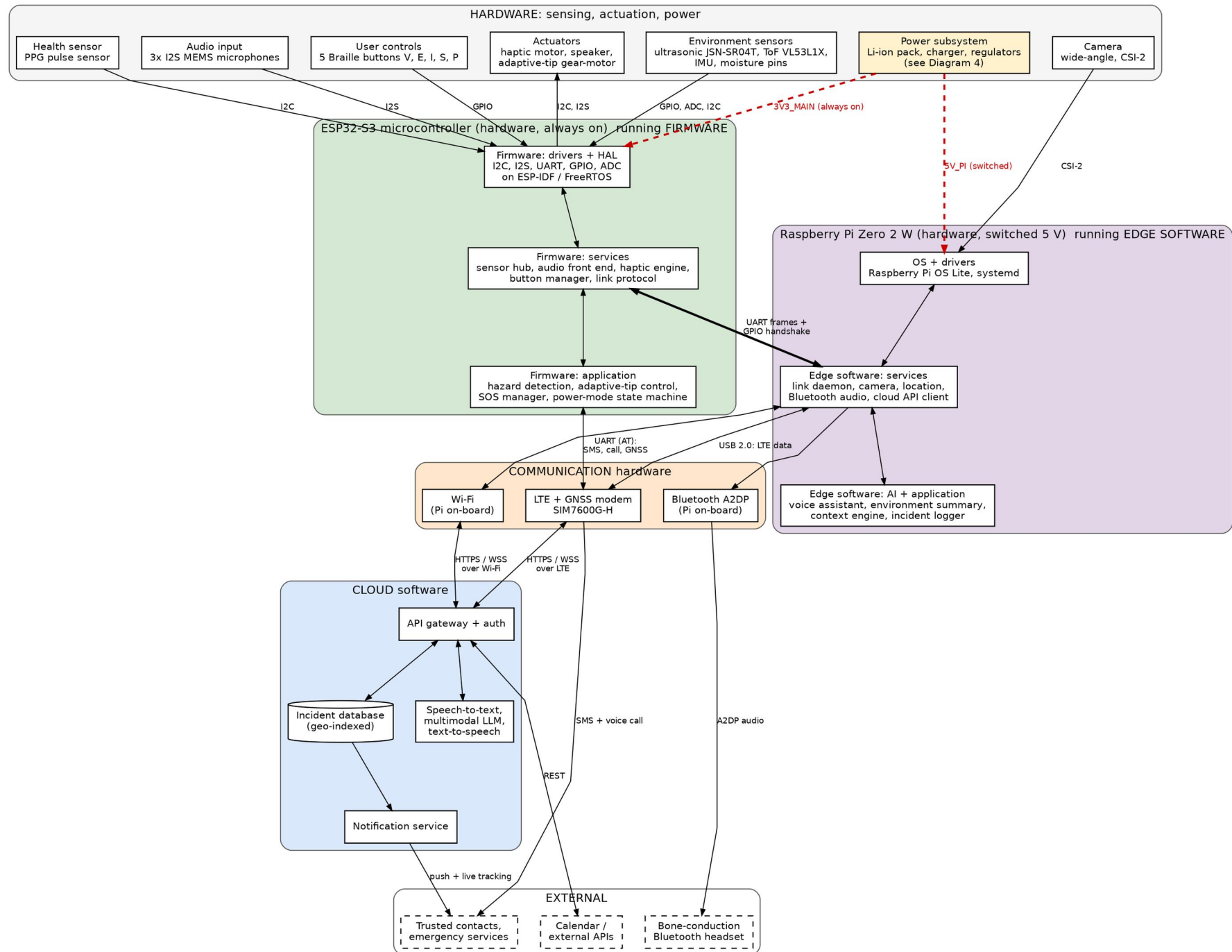
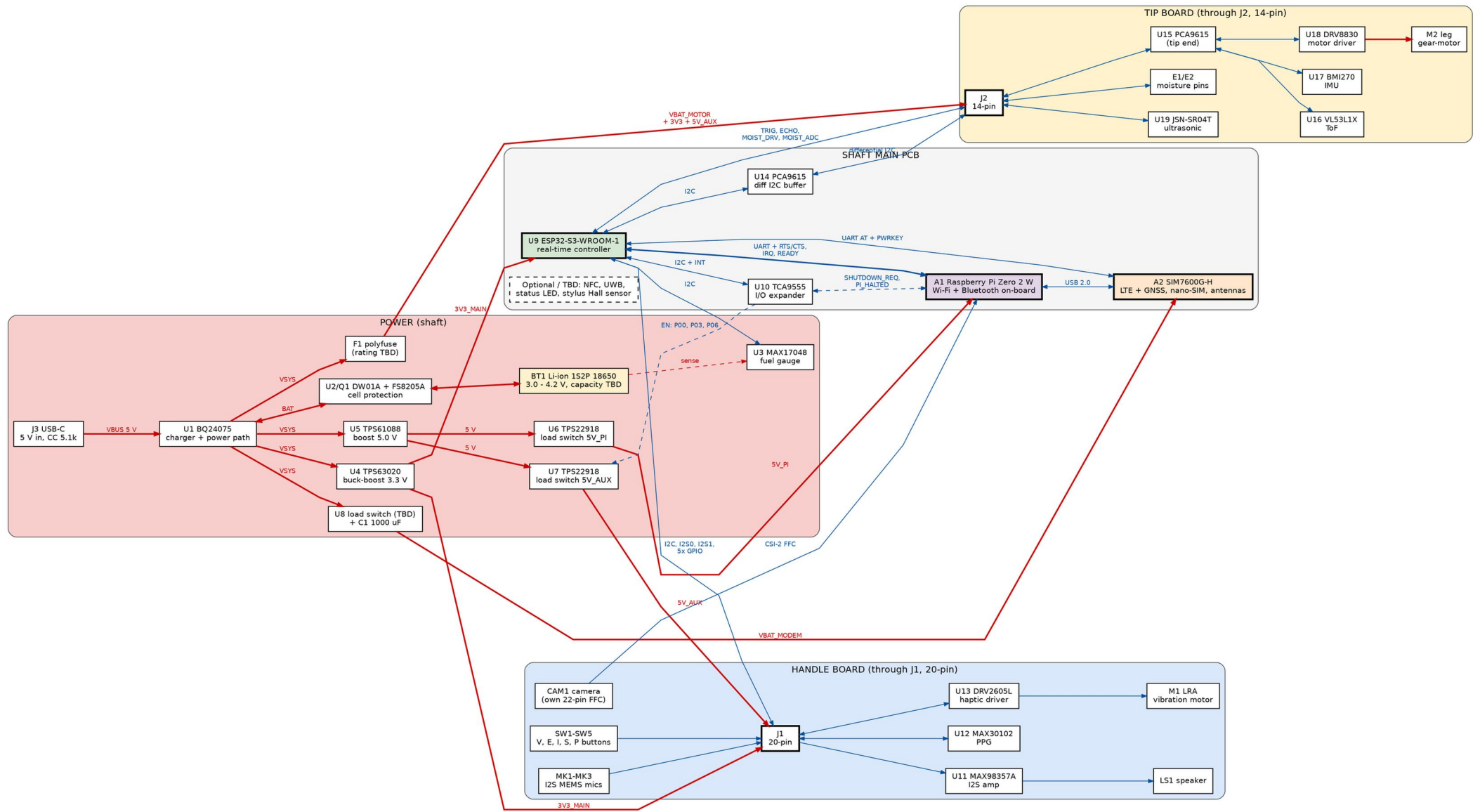
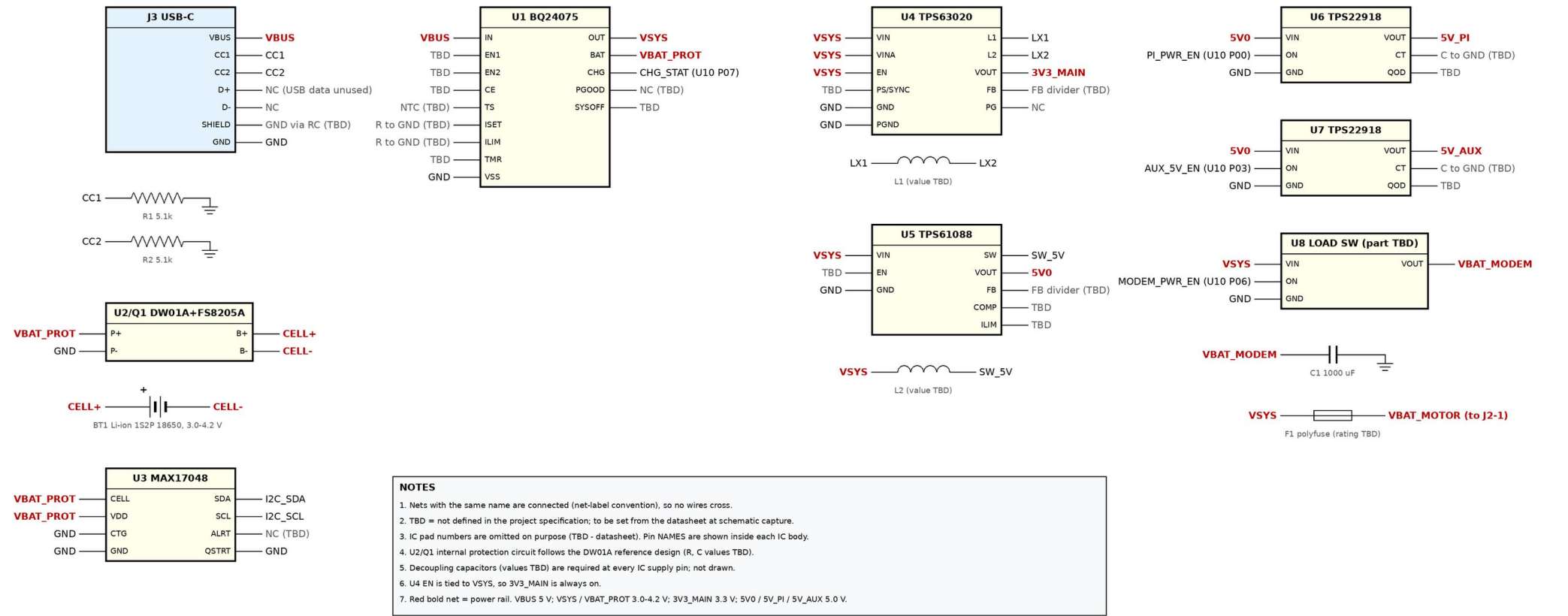


DIAGRAM 2 - HARDWARE BLOCK DIAGRAM red = POWER, blue = SIGNAL / DATA, dashed blue = control (Rev A, proposed)

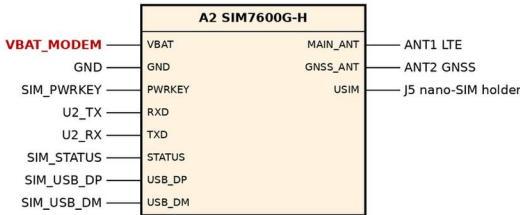
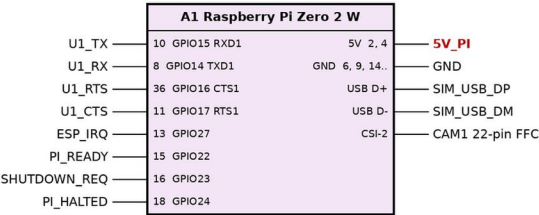
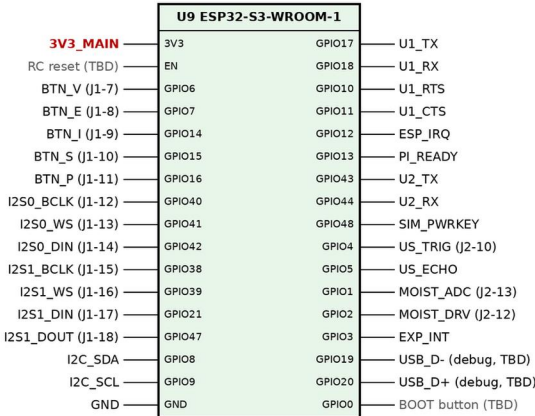
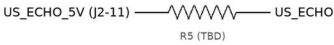
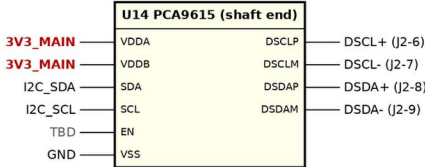
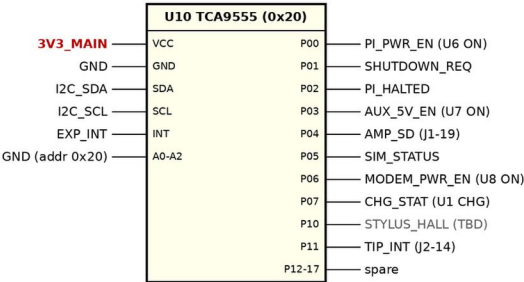


SHEET 3A - POWER INPUT, CHARGING, PROTECTION, REGULATION



Smart Intelligent Cane - Electrical schematic	
Sheet 3A: Power	Rev A
Status: PROPOSED, NOT VERIFIED	2026-09-20

SHEET 3B - CONTROLLERS, INTER-PROCESSOR LINK, MODEM, I/O EXPANDER (shaft main PCB)



NOTES

- Same net name = connected. (Jx-n) = continues on connector Jx pin n, sheet 3C.
- UART crossover: ESP TX (U1_TX) goes to Pi RXD1 pin 10; Pi TXD1 pin 8 goes to ESP RX (U1_RX). RTS / CTS are crossed the same way.
- SIM7600G-H UART / PWRKEY logic level vs 3.3 V: level translation TBD (bare module is not 3.3 V logic; many breakout boards translate on-board).
- R5 / R6 form the 5 V to 3.3 V ECHO divider; values TBD. R3 / R4 are the I2C pull-ups.
- ESP32-S3 module pad numbers TBD (datasheet). GPIO26-37, 45, 46 are unused by design.
- Pi numbers are physical 40-pin header pins. Pi USB data pads: TBD (OTG port or test pads).

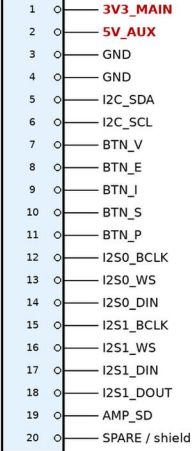
Smart Intelligent Cane - Electrical schematic

Sheet 3B: Controllers	Rev A
Status: PROPOSED, NOT VERIFIED	2026-09-20

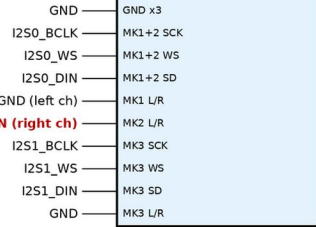
SHEET 3C - HANDLE BOARD (J1) AND TIP BOARD (J2)

HANDLE BOARD (blue parts)

J1 handle joint, 20-pin

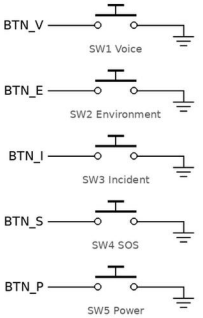
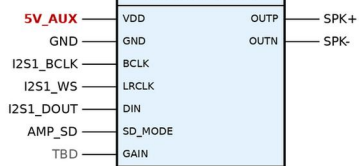


3V3_MAIN

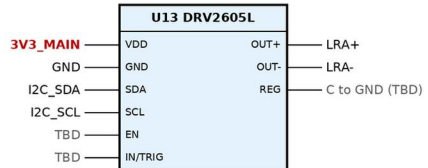
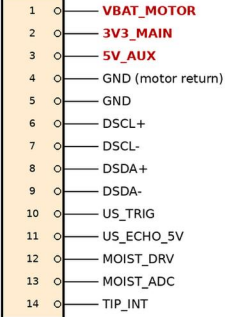


3V3_MAIN (right ch)

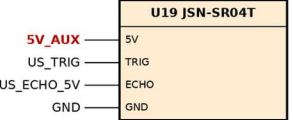
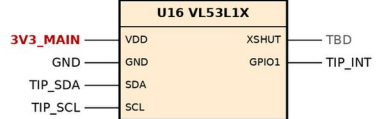
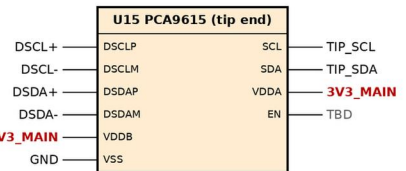
5V_AUX



J2 tip quick-release, 14-pin

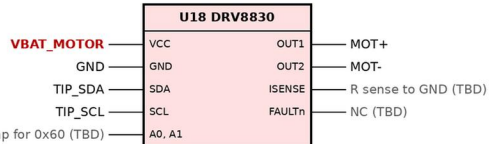
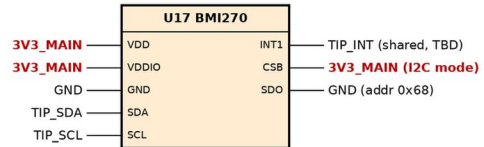


TIP BOARD (orange parts)



NOTES

1. Same net name = connected. J1 / J2 nets continue on sheet 3B.
2. Buttons are active low; pull-ups are the ESP32-S3 internal pull-ups. Debounce in firmware.
3. MAX30102 needs a 1.8 V core supply; a module with an on-board regulator is assumed (TBD).
4. Microphone part number TBD (I2S MEMS). The L/R strap selects the I2S channel.
5. TVS / ESD arrays on all J1 and J2 signal pins: part TBD, not drawn.
6. Camera CAM1 is not on J1: separate 22-pin CSI-2 PFC to A1 (sheet 3B).
7. DRV8830 (red) carries motor current from VBAT_MOTOR.



Smart Intelligent Cane - Electrical schematic	
Sheet 3C: Handle + tip	Rev A
Status: PROPOSED, NOT VERIFIED	2026-09-20

DIAGRAM 4 - POWER DISTRIBUTION ARCHITECTURE (Rev A, proposed)

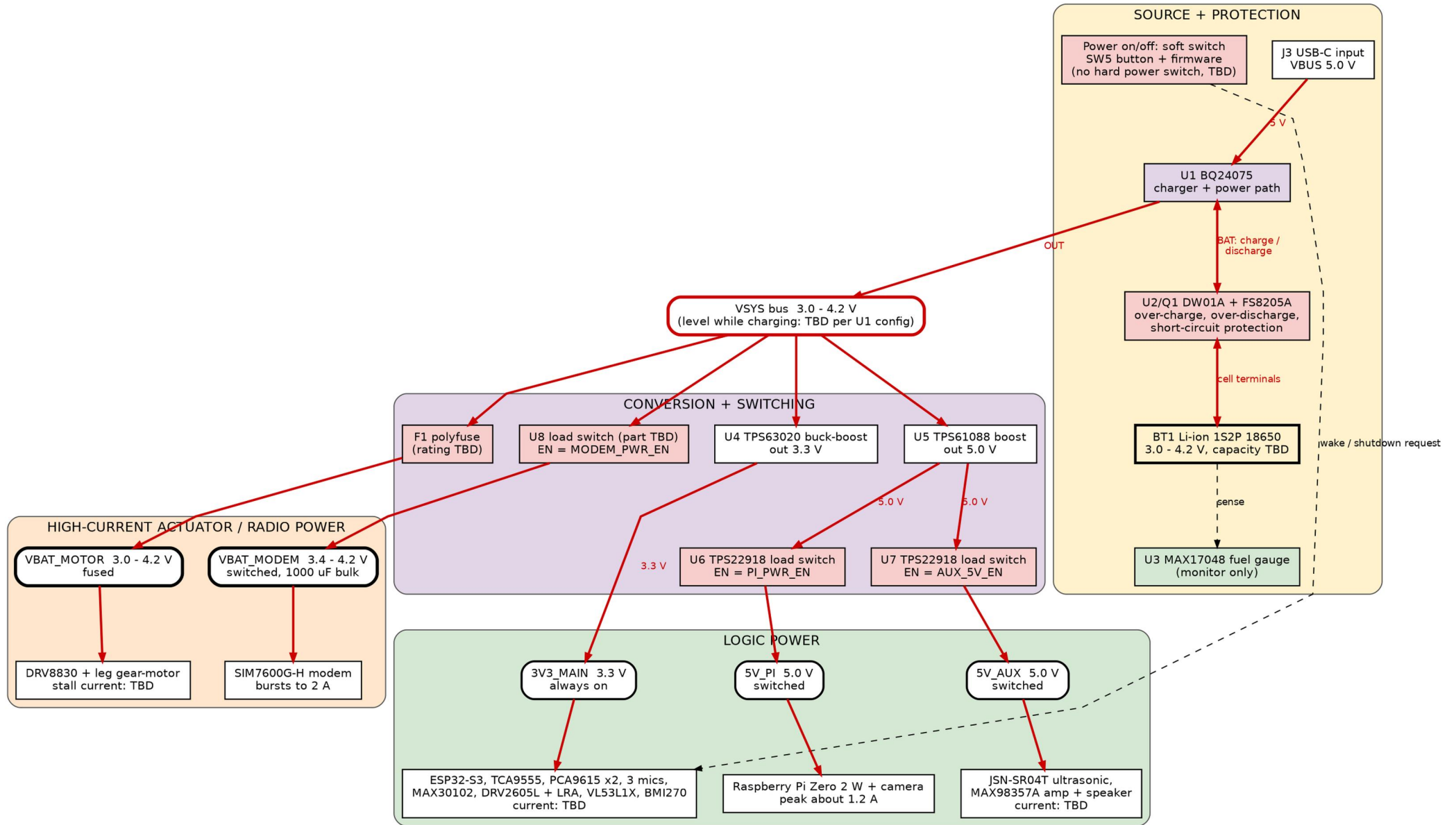


DIAGRAM 5 - HARDWARE / SOFTWARE INTERFACE (Rev A, proposed)

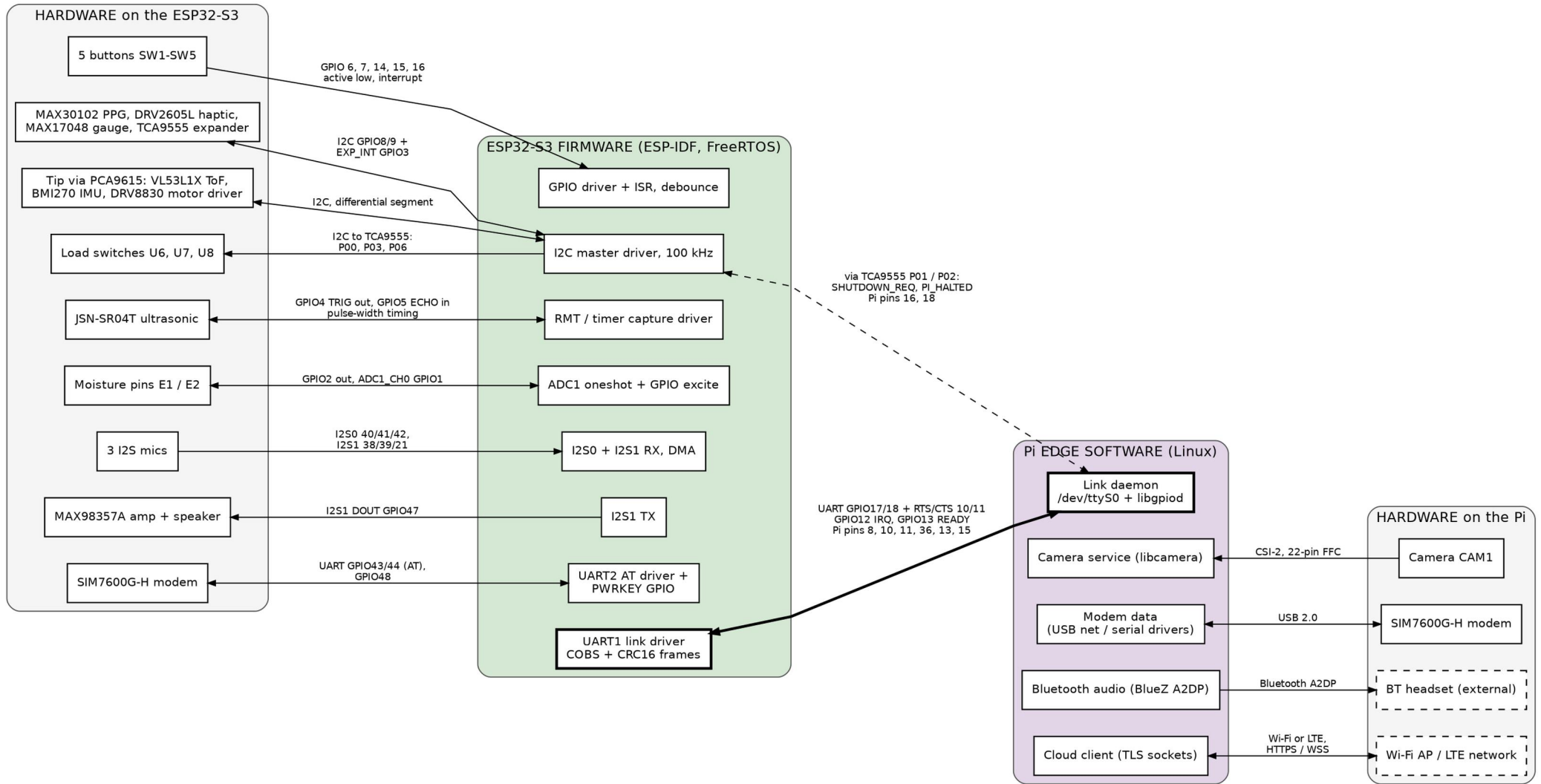


DIAGRAM 6 - SOFTWARE ARCHITECTURE (Rev A, proposed)

